

Nathan Garzya Santoso

Software Engineer

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📍 Surabaya, Indonesia

SUMMARY

Backend Software Engineer with 1.5 years of experience in designing, developing, and maintaining scalable and reliable server-side systems. Strong interest in Software Engineering principles, system architecture, and backend best practices.

Experienced in building and optimizing APIs, managing databases, handling system integrations, and designing clean backend architectures for web applications. Passionate about writing maintainable code, improving system performance, and ensuring reliability in production environments.

Continuously learning modern backend frameworks, system design patterns, and scalable architecture approaches to grow as a backend engineer and contribute to impactful, real-world projects.

EXPERIENCE

Software Engineer, Backend

Philip Morris International ↗

06/2026 - Present

Building AYO B2B Ecosystems for KSA (Kingdom of Saudi Arabia)

- Contributed to the development and maintenance of AYO Ecosystem, focusing on AYO Principal, AYO Mitra, AYO Mitra Order and AYO Toko.
- Designed and optimized complex database queries using Composite Indexes, CTEs, and Window Functions to improve performance and maintainability.
- Expanded the pricing management system from Cash-only to support both cash and credit transactions, while implementing price-based product catalogs and e2e payment flows for OTC and online transactions.
- Implemented an e2e credit payment solution, enabling customer-specific credit limits, payment terms, and flexible configurations, along with Credit Ledger Reporting, Partial and Full payment processing, and outstanding payment collection across customer and rider applications.

Software Engineer Intern

Philip Morris International ↗

12/2025 - 05/2026

Working as part of B2B Backend team.

- Contributed actively across 4 agile sprints by completing 50+ tickets, including feature implementation, bug fixes, and security improvements.
- Leveraged AWS and Jenkins to streamline deployment workflows, enabling reliable and efficient delivery of backend services.
- Migrated the authentication mechanism to a new one, more secure system to improve overall platform security. The implementation was rolled out across multiple platforms and deployed in several countries, including Indonesia, Philippines, South Africa, and Saudi Arabia.
- Integrated internal system with Microsoft Entra ID to support Single Sign-On (SSO) and automated user provisioning.
- Remediated penetration testing findings, improving overall system security and reducing potential vulnerabilities.
- Designed and wrote unit tests to ensure the reliability and correctness of new API endpoints.
- Contributed to the development and maintenance of AYO Ecosystem, focusing on AYO Principal, AYO Mitra, and AYO Toko.

Software Engineer Intern

PT Salam Pacific Indonesia Lines ↗

08/2025 - 12/2025

Developing web-based software to streamline company processes using various tech stacks.

Projects

- Analytics Dashboard: Monitoring relocation prices across all branches owned by PT Salam Pacific Indonesia Lines. Helped reduce manual work and significantly improved operational efficiency.

- Automated Performance Review Analysis: Reducing manual workload by integrating a Large Language Model to analyze employee coaching data and generate meaningful insights. The system provided strategic HR perspectives, recommended HR-level interventions such as training, program, policies, or reinforcement, and automatically classified coaching case into 5 categories for summary reporting.
- Centralized Surveillance: Built a backend system to support an AI-based anomaly detection platform for real-time CCTV monitoring. Developed RESTful API in Golang using the Service-Repository pattern, implemented real-time streaming via RTSP, FFmpeg, and WebSocket, and integrated MinIO for video storage. Designed an Event-Driven Architecture using RabbitMQ to enable real-time communication with the AI service, and implemented automated report generation with go-cron and goroutines.

Assistant Lecturer
Universitas Surabaya ↗

02/2024 - 01/2025

- Delivered engaging lectures and tutorials, ensuring student understanding of core concepts.
- Provided academic support to students through one-on-one consultations and group sessions.
- Supervised student projects, ensuring quality output.

EDUCATION

Informatics Engineering ↗
Universitas Surabaya - Bachelor of Computing

08/2022 - 04/2026

- GPA 3.974 / 4.0
- 1'st Place Database Student Challenge 2023/2024.
- 3'rd Place Object Oriented Programming Student Challenge 2022/2023.
- 1'st Place Algorithm and Programming Student Challenge 2022/2023.
- Gemastik XV and XVI Competitive Programming Participant.
- Participated in various UBAYA committees such as Maniac, MOB Universitas, ILPC, MOBFT, Industrial Games, and has been a member/coordinator of IT Division.

CERTIFICATES

Software Engineer Essentials ↗
Philip Morris International

01/2026 - Present

Pemrograman Golang : Pemula sampai Mahir ↗
Udemy

09/2025 - Present

SKILLS

PHP	Advanced	PostgreSQL	Intermediate
Golang	Intermediate	Laravel	Advanced
Python	Intermediate	Docker	Intermediate
TypeScript	Intermediate	Git	Intermediate
C#	Intermediate	AWS	Basic
MySQL	Intermediate	Redis	Basic

LANGUAGES

Indonesia	Native	English	Intermediate
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